CAFFEINE & CADAVER
MBBS BIOCHEMISTRY NOTES SERIES

PYQ 2010–2025

MBBS BIOCHEMISTRY · MODULE 03

Lipid Metabolism

β-Oxidation · Ketone Bodies · Cholesterol · Lipoproteins · Fatty Acid Synthesis

SUBJECT	EXAM RELEVANCE	PYQ FREQUENCY
Biochemistry — Paper 1	Group A · B · C · D	10+ times (2010–2025)

⭐ MBBS PYQ APPEARANCES — YEARS

2010 | 2010-S | 2011-S | 2012-S | 2013 | 2013-S | 2014 | 2015 | 2015-S | 2017 | 2017-S | 2018 | 2019 | 2019-S | 2021 | 2022 | 2023 | 2024 | 2025

📊 EXAM PRIORITY MAP

To optimize your final hours, focus on high-yield core concepts first. Use this map to navigate the module:

🔴 **TIER 1 (Must-Know Core):** Section 02 (β-Oxidation), Section 03 (Ketone Bodies), Section 05 (Cholesterol), Section 06 (Lipoproteins), Section 13 (EQ Template)

🟡 **TIER 2 (High-Yield Support):** Section 01 (Big Picture), Section 04 (Fatty Acid Synthesis), Section 07 (Phospholipids), Section 09 (Fatty Liver), Section 11 (Clinical), Section 12 (Viva Q&A)

🟢 **TIER 3 (Low-Yield / Detail-Heavy):** Section 08 (Eicosanoids), Section 10 (Mnemonics)

00: 🔥 Last Day Revision Mode

- **Filter the Noise:** Low-yield 🟢 TIER 3 sections have been collapsed into interactive dropdowns. Do not open them unless you have fully mastered Tier 1 and Tier 2.
- **Focus on Traps:** Pay special attention to the ⚠️ EXAM TRAP highlights injected into key sections — these address common points of confusion where students lose marks.
- **High-Frequency First:** Prioritize 🔴 TIER 1 topics which have appeared 10+ times in MBBS exams over the last 15 years.

01 The Big Picture 🟡 TIER 2

🩺 WHY THIS MATTERS CLINICALLY

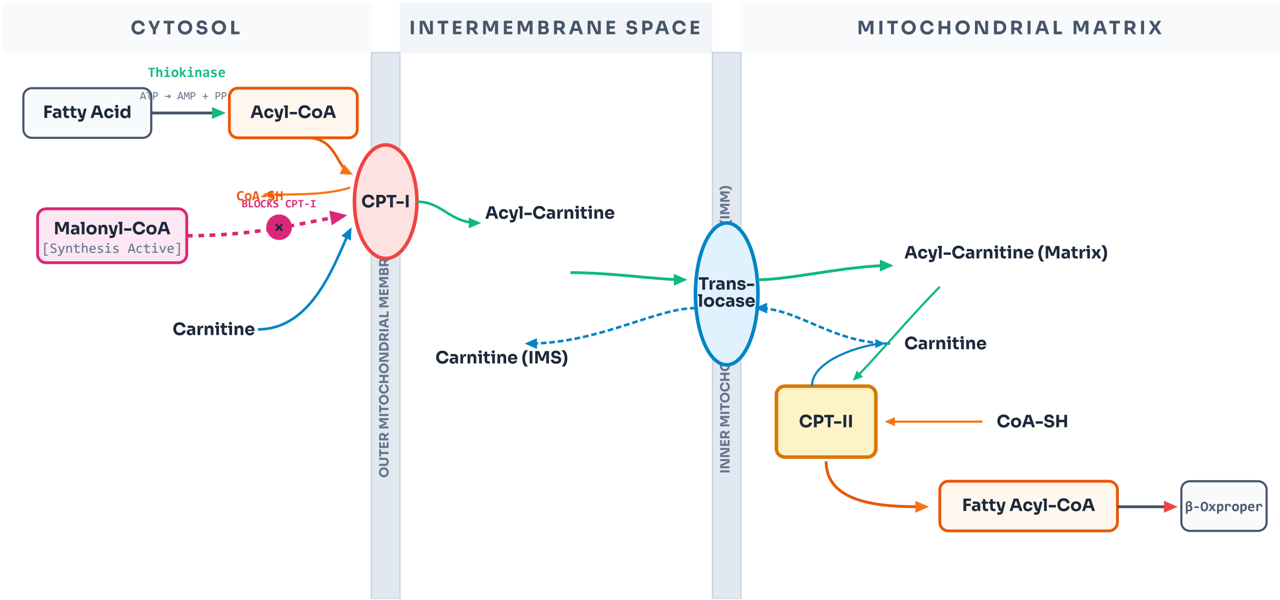
Lipid metabolism is the body's primary energy reservoir and membrane building engine, coordinating fatty acid oxidation, lipogenesis, cholesterol synthesis, and lipoprotein transport. Defects in these highly regulated pathways underlie major clinical pathologies including Diabetic Ketoacidosis (DKA), Fatty Liver Disease, atherosclerosis, and Familial Hypercholesterolaemia. Under fasting conditions, cellular traffic shifts to fat breakdown and ketone body export, while the fed state promotes de novo lipogenesis regulated by reciprocal enzyme locks.

02 β-Oxidation of Fatty Acids ★ EXAMINER FAV 🔴 TIER 1

📌 ONE-LINER

Sequential removal of **2-carbon units** as Acetyl-CoA from the β-carbon of a fatty acid, occurring in the **mitochondrial matrix**, generating $FADH_2$ + NADH per cycle.

FIG 3.1: TEXTBOOK-GRADE CARNITINE SHUTTLE & REGULATION BY MALONYL-COA



⭐ **CPT-I is the rate-limiting step** RLS of β-oxidation. Inhibited by **Malonyl-CoA** (the first committed intermediate of fatty acid synthesis) — this is how the cell prevents simultaneous synthesis and breakdown of fatty acids.

⚠️ **EXAM TRAP:** Do not confuse CPT-I and CPT-II. **CPT-I** sits on the outer mitochondrial membrane, acts as the rate-limiting gatekeeper, and is inhibited by Malonyl-CoA. **CPT-II** sits on the inner membrane and is not regulated by Malonyl-CoA.

// B-OXIDATION SPIRAL (LYNEN'S SPIRAL) CENTERED FLOWCHAIN

Fatty Acyl-CoA (n carbons)

↓ Acyl-CoA Dehydrogenase (FAD → $FADH_2$ [+1.5 ATP])

Trans-Δ^2-Enoyl-CoA

↓ Enoyl-CoA Hydratase (+ H^2O)

L-3-Hydroxyacyl-CoA

↓ 3-Hydroxyacyl-CoA Dehydrogenase (NAD^+ → NADH [+2.5 ATP])

3-Ketoacyl-CoA

↓ Thiolase (+ CoASH)

Acetyl-CoA (2C) → enters TCA cycle	**Acyl-CoA (n-2 C)** → enters next round

Palmitate (C16):
7 rounds of β-oxidation
8 Acetyl-CoA produced
Per round:
1 $FADH_2$ + 1 NADH
= 4 ATP
Net ATP (C16):
106 ATP total
(−2 for activation)

🧠 MNEMONIC — B-OXIDATION 4 STEPS

"OHOT" → Oh, How Oxidation Triumphs!

Oxidation (Acyl-CoA Dehydrogenase) → **H**ydration (Enoyl-CoA Hydratase) → **O**xidation (3-Hydroxyacyl-CoA Dehydrogenase) → **T**hiolysis (Thiolase).

SPECIAL OXIDATIONS

α-Oxidation: Oxidation at α-carbon. For branched-chain fatty acids (e.g. phytanic acid). Requires Phytanoyl-CoA hydroxylase. Defect → **Refsum's disease**.
ω-Oxidation: Oxidation at terminal (ω) carbon. Minor pathway in ER. Produces dicarboxylic acids. Important in MCAD deficiency.

Peroxisomal β-Oxidation: For very long chain FA (>C20). Generates H^2O^2 (catalase destroys it). No ATP produced directly. Defect → **Zellweger syndrome**, **Adrenoleukodystrophy (ALD)**.

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03 Ketone Bodies ★ EXAMINER FAV 🔴 TIER 1

// KETOGENESIS & KETOLYSIS RECIPROCAL CENTERED FLOWS

LIVER (SYNTHESIS ONLY)

2 × Acetyl-CoA

↓ Thiolase

Acetoacetyl-CoA

↓ HMG-CoA Synthase RLS

HMG-CoA

↓ HMG-CoA Lyase

Acetoacetate (1° KB)

↓ spontaneous / NADH red^n

β-Hydroxybutyrate / Acetone

EXTRAHEPATIC (UTILIZATION)

β-Hydroxybutyrate

↓ Dehydrogenase

Acetoacetate

↓ **Thiophorase[ABSENT IN LIVER!]**

Acetoacetyl-CoA

↓ Thiolase

2 × Acetyl-CoA → TCA

🧠 MNEMONIC — KETONE BODY NAMES

"A Beta-Blocker Acetates" → ABA

Acetoacetate · **B**eta-hydroxybutyrate · **A**cetone. Only the first two carry energy; acetone is exhaled (fruity breath in DKA).

🏭 **Ketone bodies are made in LIVER but used in EXTRAHEPATIC tissues** (brain, heart, muscle, kidney). Liver lacks Succinyl-CoA transferase (thiophorase) — so it cannot use its own ketones.

🔄 **Ketolysis (utilisation) — in extrahepatic tissues:** β-Hydroxybutyrate → Acetoacetate (by HBD) → Acetoacetyl-CoA (by Succinyl-CoA transferase, needs Succinyl-CoA) → 2 Acetyl-CoA → TCA → ATP.

🩺 Ketosis — Clinical Scenarios

- **Starvation ketosis:** Mild — OAA diverted to GNG → Acetyl-CoA cannot enter TCA → ketogenesis. Controlled, compensated. pH usually normal.
- **DKA (Diabetic Ketoacidosis):** Severe — No insulin → uncontrolled lipolysis → massive FFA → overwhelming ketogenesis. Acetone breath, pH < 7.3, High Anion Gap Metabolic Acidosis. Emergency.
- **Alcoholic ketoacidosis:** Ethanol → NADH ↑ → OAA → Malate (depletes OAA) → same result as starvation.

⚠️ **EXAM TRAP:** Both starvation AND DKA cause ketosis, but **DKA is far more severe** because absolute insulin deficiency completely removes the brake on lipolysis, leading to massive, uncontrolled FFA release and life-threatening high anion gap metabolic acidosis.

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04 Fatty Acid Synthesis (Lipogenesis) TIER 2

// DE NOVO FATTY ACID SYNTHESIS — KEY STEPS

Acetyl-CoA (mitochondria) → **Citrate shuttle to cytoplasm** Acetyl-CoA (cytoplasm)

↓ Acetyl-CoA Carboxylase (ACC) RLS + Biotin + CO_2 + ATP

Malonyl-CoA (3C)

↓ Fatty Acid Synthase Complex (FAS) — 7 rounds × 2C addition

Palmitate (C16:0) — first product

Location:
Cytoplasm
(+ SER for elongation)
Reducing agent:
NADPH (from HMP shunt
+ malic enzyme)
Cost per Palmitate:
7 ATP + 14 NADPH
+ 8 Acetyl-CoA

⭐ **ACC (Acetyl-CoA Carboxylase)** RLS — Activated by: Citrate, Insulin. Inhibited by: Palmitoyl-CoA (product inhibition), Glucagon, Epinephrine (via phosphorylation by PKA). **Malonyl-CoA** — the product — also inhibits CPT-I, preventing simultaneous FA synthesis + breakdown.

⚠️ **EXAM TRAP:** Fatty acid synthesis occurs in the **cytoplasm**, but Acetyl-CoA is generated in the **mitochondria**. Remember that Acetyl-CoA cannot cross the inner mitochondrial membrane directly; it must be converted to **Citrate** to enter the cytoplasm via the **Citrate Shuttle**.

Synthesis vs β-Oxidation

Location: **Cytoplasm** vs **Mitochondria**

Carrier: **ACP (Synthesis)** vs **CoA (Oxidation)**

NADPH consumed vs NADH+$FADH_2$ produced

Citrate shuttle IN vs Carnitine shuttle IN

Stereochemistry: **L** vs **D** intermediate

Elongation & Desaturation

Elongation: ER + mitochondria (Malonyl-CoA units)

Desaturation: ER → Δ9-desaturase (most common)

Humans: **cannot introduce double bonds beyond C9**

∴ Linoleic (ω6) + α-Linolenic (ω3) = **Essential FA**

EFA deficiency → scaly skin, ↑ susceptibility to infection

05 Cholesterol Synthesis ★ EXAMINER FAV 🔴 TIER 1

// CHOLESTEROL SYNTHESIS & STATIN REGULATION CENTERED FLOW

3 × Acetyl-CoA

↓ HMG-CoA Synthase

HMG-CoA

↓ HMG-CoA Reductase RLS

× Competitive Inhibition by Statins

Mevalonate (Statin block target)

↓ IPs → Squalene → Lanosterol

Cholesterol (27C)

Location:
Cytoplasm + ER
(mainly liver)
ACAT:
Esterifies cholesterol
for storage
LCAT:
Esterifies cholesterol
in plasma (HDL)

💊 **HMG-CoA Reductase** RLS — Inhibited by: Statins (competitive), High intracellular cholesterol (via SREBP-2 pathway), Glucagon. Activated by: Insulin. **Statins** (Atorvastatin, Rosuvastatin) → ↓ cholesterol → upregulate LDL receptors → ↑ LDL clearance from blood.

⚠️ **EXAM TRAP:** Statins are **competitive inhibitors** of HMG-CoA Reductase. They do not destroy the enzyme; they compete with HMG-CoA. This leads to a compensatory upregulation of LDL receptors on the hepatocyte surface, which is the actual mechanism that lowers circulating LDL.

PRODUCTS OF CHOLESTEROL

1Bile acids (primary: cholic, chenodeoxycholic) — essential for fat digestion
2Steroid hormones — glucocorticoids, mineralocorticoids, sex hormones
3Vitamin D — 7-dehydrocholesterol → D3 (skin, UV light)
4Cell membrane — modulates fluidity

06 Plasma Lipoproteins ★ EXAMINER FAV 🔴 TIER 1

LIPOPROTEIN	ORIGIN	MAIN LIPID	KEY APO	FUNCTION
CHYLOMICRON	Intestine	TAG (85%)	ApoB-48, ApoC-II, ApoE	Transport dietary TAG to tissues
VLDL	Liver	TAG (55%)	ApoB-100, ApoC-II, ApoE	Transport endogenous TAG
IDL	VLDL remnant	TAG + Chol	ApoB-100, ApoE	Intermediate; → LDL
LDL	IDL (VLDL remnant)	Cholesterol (45%)	ApoB-100	**Delivers cholesterol to cells — "Bad"**
HDL	Liver + Intestine	Protein (50%)	ApoA-I, ApoA-II	**Reverse cholesterol transport — "Good"**

🧠 MNEMONIC — LIPOPROTEIN DENSITY ORDER

"Chubby Vikings Inspire Lovely Humans"

Chylomicron → **V**LDL → **I**DL → **L**DL → **H**DL. Density increases (↑ protein, ↓ fat): Chylomicrons are least dense, HDL is most dense.

🧠 MNEMONIC — APOLIPOPROTEIN FUNCTIONS

"A-I Activates; B-100 Binds; C-II Calls; E Everyone enters"

ApoA-I → Activates LCAT (HDL) · **ApoB-100** → Binds LDL receptor · **ApoC-II** → Activates LPL (TAG delivery) · **ApoE** → Mediates remnant uptake.

// VLDL METABOLISM & REVERSE CHOLESTEROL TRANSPORT

VLDL (from liver) → **LPL (capillaries)** → Releases TAG → tissues

↓ becomes IDL → LPL again

LDL → **LDL receptor (ApoB-100)** → Cell takes up cholesterol

HDL (nascent) from liver/intestine → **LCAT (ApoA-I activates)** → Picks up free cholesterol from tissues

↓

Mature HDL → **CETP** → Cholesterol → Liver (for excretion as bile)

⚖️ **LPL vs. HSL (Classic Viva Question):** Lipoprotein Lipase (LPL) sits on capillaries, is activated by Insulin and ApoC-II, and clears TAGs from the blood into tissues (Fed state). Hormone Sensitive Lipase (HSL) sits inside adipocytes, is activated by Glucagon/Epinephrine, and breaks down stored fat to release free fatty acids into the blood (Fasting state).

🧬 **Familial Hypercholesterolaemia (FH):** LDL receptor defect (ApoB-100 mutation) → LDL not cleared → ↑ serum LDL → premature atherosclerosis + xanthomas. Heterozygous: LDL ~2× normal. Homozygous: LDL ~6× normal, MI before age 20.

⚠️ **EXAM TRAP:** Remember that **ApoB-48** is unique to **Chylomicrons** (intestinal origin, representing 48% of the ApoB gene due to RNA editing), while **ApoB-100** is found on **VLDL, IDL, and LDL** (hepatic origin, 100% full-length translation).

07 Phospholipids & Glycolipids TIER 2

Core Phospholipid & Glycolipid Pathologies — High Yield

- **Lung Surfactant (DPPC / Dipalmitoyl Lecithin):** Secreted by Type II pneumocytes. Reduces alveolar surface tension → prevents collapse. Prematurity (<35 weeks) → surfactant deficiency → **Respiratory Distress Syndrome (RDS)**. Amniotic **L:S ratio > 2** indicates lung maturity.
- **Tay-Sachs Disease: Hexosaminidase A** deficiency → **GM2 Ganglioside** accumulation. Leads to neurodegeneration, developmental delay, and **cherry-red spot on macula** (NO hepatosplenomegaly).
- **Gaucher's Disease (Most Common): Glucocerebrosidase** deficiency → **Glucocerebroside** accumulation. Leads to hepatosplenomegaly, bone crises, and characteristic **"crumpled tissue paper" macrophages**.
- **Niemann-Pick Disease: Sphingomyelinase** deficiency → **Sphingomyelin** accumulation. Leads to neurodegeneration, **cherry-red spot on macula**, AND **hepatosplenomegaly** (foam cells).

08 Eicosanoids TIER 3

Arachidonic Acid Cascade & Pharmacology

Precursor: Arachidonic Acid (C20:4) is released from membrane phospholipids by **Phospholipase A_2 (PLA_2)** (which is inhibited by anti-inflammatory corticosteroids).

COX Pathway: Converts arachidonic acid to **Prostaglandins (PGE_2, PGI_2 / Prostacyclin)** (vasodilation, renal blood flow) and **Thromboxanes (TXA_2)** (vasoconstriction, platelet aggregation).

LOX Pathway: Converts arachidonic acid to **Leukotrienes (LTA_4, LTB_4, LTC_4)** (major bronchoconstrictors and mediators of allergic asthma).

COX Inhibition:

- **Aspirin (Irreversible):** Irreversibly acetylates COX (serine residue). Anti-platelet effect persists for **7–10 days** (platelet lifespan) because platelets lack a nucleus and cannot synthesize new COX.
- **NSAIDs (Reversible):** Reversible competitive inhibitors (e.g., Ibuprofen).
- **Celecoxib (Selective):** Selective COX-2 inhibitor (spares COX-1 to protect gastric mucosa, reducing ulcer risk).

09 Fatty Liver & Lipotropic Factors TIER 2

🩺 Hepatic Steatosis (Fatty Liver) & Lipotropic Factors

Definition: Accumulation of TAG in hepatocytes >5% of liver weight, resulting from an imbalance between liver TAG synthesis/intake and VLDL export.

- **Causes of Accumulation:**
  - **↑ FA Delivery:** Starvation, uncontrolled Diabetes Mellitus, obesity.
  - **↑ FA Synthesis:** High carbohydrate diet or high insulin levels.
  - **↓ FA Oxidation:** Alcohol ingestion (high $NADH/NAD^+$ blocks oxidation) or hypoxia.
  - **↓ VLDL Secretion:** Carbon tetrachloride (CCl_4) poisoning, protein malnutrition (Kwashiorkor) depletes apoproteins.
- **Lipotropic Factors:** Substances required for the synthesis and export of VLDL, thereby preventing lipid accumulation in the liver. Key factors include **Choline** (constituent of phosphatidylcholine in VLDL membrane) and **Methionine** (methyl donor for choline synthesis).

🧠 MNEMONIC — LIPOTROPIC FACTORS

"Clever Mice In Big Labs"

Choline · **M**ethionine · **I**nositol · **B**12 · **L**ecithin. Deficiency of any of these blocks VLDL export → Fatty Liver.

🔥 VIVA RAPID FIRE — LIPID METABOLISM

Q1 Why can't even-chain fatty acids contribute to net gluconeogenesis?

Even-chain FA → β-oxidation → Acetyl-CoA only. Acetyl-CoA cannot be converted back to pyruvate or OAA in net amounts because the Pyruvate Dehydrogenase reaction is irreversible. Acetyl-CoA entering TCA condenses with OAA → both carbons are released as CO_2 in the cycle — no net carbon gain for GNG. Exception: odd-chain FA → Propionyl-CoA → Succinyl-CoA → can enter GNG.

Q2 Why does the liver produce but not use ketone bodies?

Liver lacks Succinyl-CoA transferase (3-oxoacid CoA transferase / thiophorase) — the enzyme needed for the first step of ketolysis. This is actually physiologically appropriate: the liver produces ketone bodies specifically to export fuel to the brain and other tissues during starvation. If the liver used its own ketones, this export function would fail.

Q3 How does Malonyl-CoA coordinate FA synthesis and oxidation?

Malonyl-CoA is the first committed product of FA synthesis (made by ACC). It simultaneously inhibits CPT-I — the gatekeeper for FA entry into mitochondria for β-oxidation. So when FA synthesis is ON (high insulin, fed state), malonyl-CoA rises and blocks β-oxidation. When fasting → glucagon → ↓ malonyl-CoA → β-oxidation proceeds. One molecule coordinates two opposing pathways.

Q4 Calculate ATP yield from complete oxidation of palmitate (C16:0).

Activation: −2 ATP (Acyl-CoA Synthetase). 7 rounds of β-oxidation: 7 $FADH_2$ (×1.5 = 10.5 ATP) + 7 NADH (×2.5 = 17.5 ATP) = 28 ATP. 8 Acetyl-CoA → TCA: 8 × 10 = 80 ATP. Total = 80 + 28 − 2 = **106 ATP** (modern P/O ratios). Using older textbook values (3 per NADH, 2 per $FADH_2$): 8×12 + 7×5 − 2 = 129 ATP. Match whichever your professor uses.

Q5 What is reverse cholesterol transport and why is HDL "good"?

Reverse cholesterol transport is the process by which HDL picks up excess free cholesterol from peripheral tissues and transports it back to the liver for excretion as bile acids. Step 1: Nascent HDL (disc-shaped) secreted by liver/intestine. Step 2: ApoA-I activates LCAT → esterifies cholesterol → HDL matures (spherical). Step 3: Mature HDL delivers cholesterol ester to liver via SR-B1 receptor OR transfers via CETP to LDL/VLDL. This removes cholesterol from artery walls → anti-atherogenic = "good."

13 Group-D EQ Template

🔴 TIER 1

📝 HOW TO ANSWER: "HDL IS GOOD CHOLESTEROL" / "STATIN DRUGS ACT AS CHOLESTEROL-LOWERING AGENTS"

Use the flowchart formula — Define → Mechanism → Kinetics/Regulation → Clinical consequence.

// EQ ANSWER FLOWCHART — "STATINS ACT AS CHOLESTEROL-LOWERING AGENTS"

Define
Competitive inhibitors of HMG-CoA reductase

↓

Mechanism
Statin → ↓ Mevalonate → ↓ Cholesterol synthesis in hepatocytes

↓

Compensatory upregulation
↓ Intracellular cholesterol → SREBP-2 activated → ↑ LDL receptor expression

↓

Clinical outcome
↑ LDL receptor → ↑ clearance of LDL from plasma → ↓ serum LDL

↓

Side effect
Myopathy (↑ CK-MM) — statin-induced myositis

MBBS Biochemistry — Module 03: Lipid Metabolism

Compiled from PYQ 2010-2025

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